Cognitive Interaction with Robots

Fall 2026 – 2027

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Course Overview

Course Overview

Course Contents

1. Introduction to Cognitive Robotics
2. Classical Robot Control
3. Reactive & Behavior-Based Robotics
4. Cognitive Architectures
5. Perception for Cognitive Robots
6. Knowledge Representation & Reasoning
7. Human–Robot Interaction Fundamentals

Course Overview

Course Learning Outcomes (CLO's)

By the end of this course, students will be able to:

1. Explain core theories of cognition and how they map onto robotic control architectures.
2. Compare classical, reactive, and hybrid robot control paradigms and articulate their trade-offs.
3. Implement perception–action loops and basic cognitive architectures on simulated or physical robots.

Course Overview

Course Learning Outcomes (CLO's)

4. Apply principles of embodied and situated cognition to robot design decisions.
5. Design and evaluate human–robot interaction scenarios grounded in social and cognitive science.
6. Critically assess the ethical, safety, and societal implications of cognitive and social robots.

Course Overview

Assessment & Grading

1.	Class Participation	**10%**
2.	Quizzes	**10%**
3.	Assignment(s) / Project	**5%**
4.	Midterm Exam	**25%**
5.	Final Exam	**50%**

Lecture 1

Introduction to Cognitive Robotics

Introduction to Cognitive Robotics

Contents

1. What is Cognitive Robotics?
2. Why do We Need Cognitive Robotics?
3. Cognitive Interaction with Robots
4. Industrial Requirements for Cognitive Robots

1.1 What Is Cognitive Robotics?

What Is Cognitive Robotics?

Cognitive Robotics

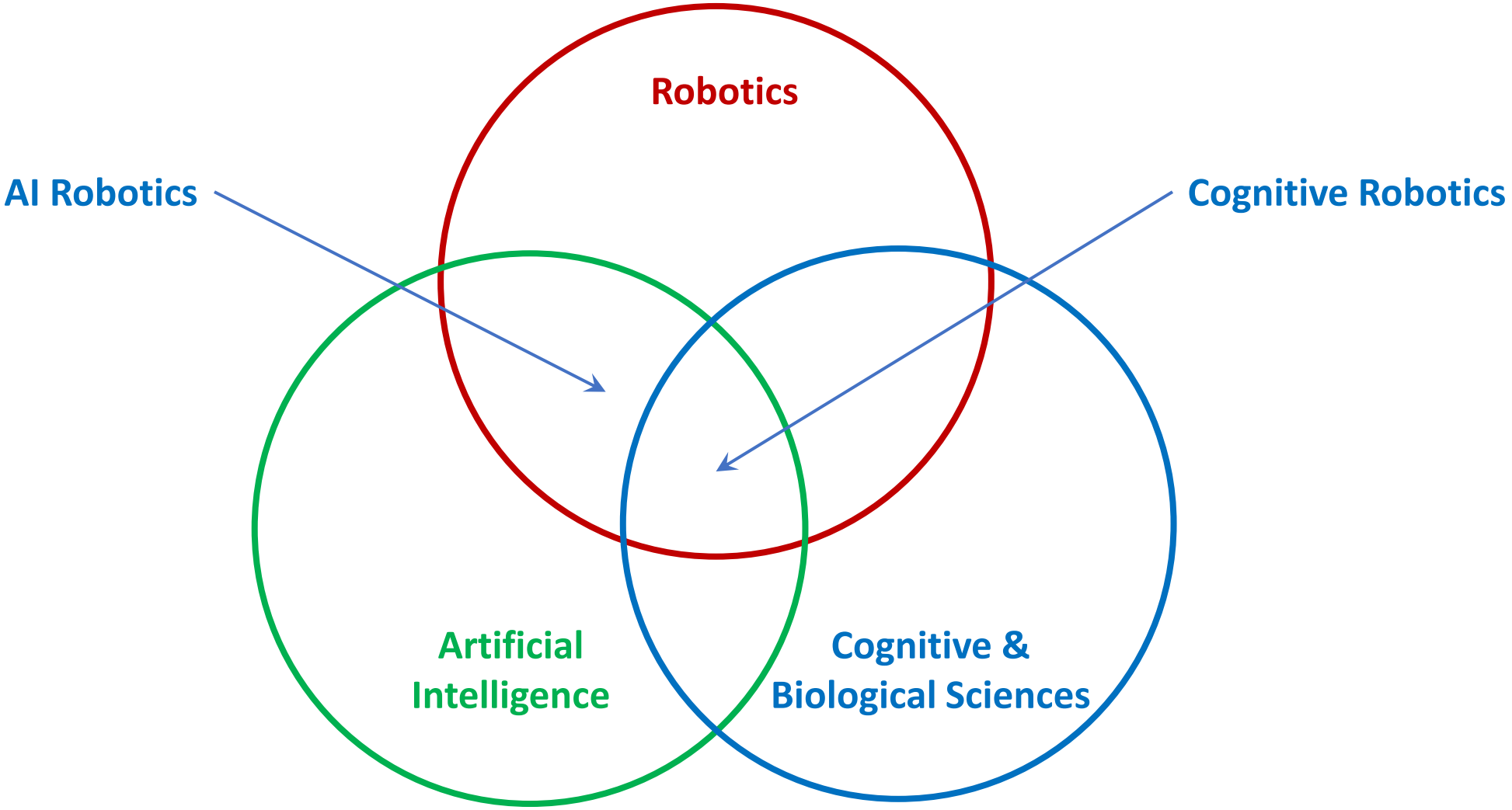
- Cognitive robotics is the study of how to build robots that perceive, reason, learn, and act intelligently in open, unstructured, and often human-populated environments — rather than simply repeating pre-programmed motions.

What Is Cognitive Robotics?

The field draws on several disciplines:

- Robotics — sensing, actuation, and control.
- Artificial Intelligence — planning, learning, and reasoning.
- Cognitive Science — models of mind and cognition.
- Psychology & Neuroscience — human behavior and perception

What Is Cognitive Robotics?



What is Cognitive Robotics

What is Cognition?

- Cognition is the process by which an autonomous system perceives its environment, learns from experience, anticipates the outcome of events, acts to pursue goals, and adapts to changing circumstances.

1.2 Why Do We Need Cognitive Robots?

Why Do We Need Cognitive Robots?

Three contrasting settings illustrate why cognition matters

- **Structured factories**

  Fixed layouts, known objects, and repeated tasks — the setting where classical robots thrive.

- **Unstructured homes and streets**

  Novel objects, changing layouts, and unpredictable events on a daily basis.

- **Human-populated spaces**

  Robots must interpret human intentions and coordinate safely with people.

Classical vs. Cognitive Robotics

Dimension	Classical Robotics	Cognitive Robotics
Control logic	Fixed, hand-engineered rules	Architectures that *reason*, *plan*, and *learn*
Environment assumption	Known, structured environment	Open, *unstructured* environment
Core cycle	**Sense → Plan → Act** (rigid pipeline)	Continuous **perception–cognition–action** loop
Adaptation	Limited adaptation to novelty	Adapts *via learning* and *generalization*
Memory & goals	Little or no learning from experience	Incorporates *memory*, *goals*, and *interaction*

The Perception–Cognition–Action Loop

A cognitive robot runs a continuous loop

- **Perceive** — sense the world through vision, audio, proprioception, and sensor fusion.
- **Cognize** — interpret, reason, plan, retrieve memory, and set goals.
- **Act** — execute motor commands to manipulate objects or navigate.

This loop is closed through the *environment*, *memory*, and *learning*, distinguishing it from the rigid, one-directional Sense–Plan–Act pipeline of classical robotics.

1.3 Cognitive Interaction with Robots

Cognitive Interaction with Robots

If a robot must operate around people, the robot must perceive, model, and communicate with humans:

- **Communication modalities** — speech, gesture, gaze, and prosody as interaction channels.
- **Social cognition** — joint attention, intention recognition, and theory of mind.

Cognitive Interaction with Robots

- **Trust & collaboration** — calibrated trust, shared autonomy, and human-in-the-loop control.
- **Ethics & safety** — anthropomorphism, privacy, and responsible deployment.

1.4 Industrial Requirements for Cognitive Robots

Industrial Requirements for Cognitive Robots

1. Safe, reliable, and transparent operation

- A cognitive robot should operate *reliably* and *safely* around humans
- It should explain the decisions it makes, the actions it has taken, and the actions it is about to take
- It should have limited autonomy to set intermediate goals to when carrying out tasks set by users

Industrial Requirements for Cognitive Robots

1. Safe, reliable, and transparent operation

- It should defer to the user's preferences apart from some exceptional circumstances, e.g. people with *dementia* can interact in unpredictable ways and the robot will be able to recognize these situations and adapt in some appropriate manner
- A cognitive robot should ask for assistance when necessary

Industrial Requirements for Cognitive Robots

1. Safe, reliable, and transparent operation

- In emergency situations, it should stop all tasks to follow some emergency procedure
- If the user is deliberately trying to misuse the robot, e.g. programming it to assist with some unethical task, a cognitive robot should cease operation

Industrial Requirements for Cognitive Robots

2. High-level instruction and context-aware task execution

- Goals and tasks for a cognitive robot can be expressed using high-level instructions using natural language and gestures

Industrial Requirements for Cognitive Robots

2. High-level instruction and context-aware task execution

- A cognitive robot should factor in contextual constraints that are specific to the application scenario when carrying out these tasks
  - Instructions can be grounded in the codified organizational rules, regulations, and behavioral guidelines that apply to a given application environment.
  - This grounding provide constraints which should make it easier for the robot to understand and perform the task effectively

Industrial Requirements for Cognitive Robots

2. High-level instruction and context-aware task execution

- A goal can be given in non-specific terms; the robot should identify the sub-goals that are needed to achieve the goal
- A cognitive robot should learn ways of measuring the success of outcomes for the goals that have been set
- It should be possible to pre-load knowledge about the robot's purpose and its operating environment, including any rules or constraints that apply to behavior in that environment

Industrial Requirements for Cognitive Robots

2. High-level instruction and context-aware task execution

- It should be possible to utilize domain-specific skill pools (e.g. from shared databases) so that the robot is pre-configured to accomplish basic tasks without having to resort to learning or development
- The level of detail in the goal description can decrease over time as the robot gains experience
- It should be possible to instruct the robot off-line if there is no access to the physical site; e.g., using a simulation tool

Industrial Requirements for Cognitive Robots

3. Knowledge Acquisition and Generalization

- A cognitive robot should continuously acquire new knowledge and generalize that knowledge so that it can undertake new tasks by generating novel action policies based on its history of decisions
  - This will allow the rigor and level of detail with which a human expresses the task specification to be relaxed on future occasions

Industrial Requirements for Cognitive Robots

3. Knowledge Acquisition and Generalization

- In general, a cognitive robot should extract useful meaning from an interaction for a future and more general use, with the same or another user
  - This may extend to learn cultural preferences and social norms

Industrial Requirements for Cognitive Robots

4. Adaptive Planning

- A cognitive robot should be able to anticipate events, prepare for them in advance, and compensate for future conditions
  - For example, an automated combine harvester should be able to apply a pre-emptive increase of power to compensate for the demands caused when an area of high yield is encountered

Industrial Requirements for Cognitive Robots

4. Adaptive Planning

- It will be able to cope with unforeseen situations, recognizing and handling errors, gracefully and effectively
  - This may include retrying with a slightly different strategy
  - The learning process will be fast, ideally learning from each error

Industrial Requirements for Cognitive Robots

4. Adaptive Planning

- A cognitive robot should be able to learn about the environment, adapt to changes in the environment, verifying that the environment matches with what is known, or there is a change and updates
  - This may require an update of the task but only after asking the user

Industrial Requirements for Cognitive Robots

4. Adaptive Planning

- A cognitive robot should be able to manipulate flexible or live objects, e.g. living creatures such as laboratory mice
  - To do so means that the robot must be able to construct a model of their behaviour and adapt its actions as required, continually refining the mode

Industrial Requirements for Cognitive Robots

5. Personalized Interaction

- A cognitive robot will personalize their interactions with humans, adapting its behavior and interaction policy to the user's preferences, needs, and emotional or psychological state
- It should learn the personal preferences of the person with whom it is interacting
  - For example, an autonomous car will learn the preferred driving style of the owner and adopt that style to engender trust

Industrial Requirements for Cognitive Robots

5. Personalized Interaction

- A cognitive robot should understand nuances in tone to learn a person's voice, detecting signs of stress so that it can react to it and review what it is doing
- In cases where showing the robot what to do involves physical contact between the user and the robot, the robot should be able to learn the dynamics of the user, i.e. his or her personal preferred use of forces when interacting with objects in the environment

Industrial Requirements for Cognitive Robots

5. Personalized Interaction

- It should be able to monitor the psychological state of a user, e.g. based on the facial expressions, gestures, actions, movements. Based on this, it should be able to determine what they need by cross-referencing that with knowledge of the person's history

Industrial Requirements for Cognitive Robots

6. Self-Assessment

- Cognitive robots should be able to reason about their own capabilities, being able to determine whether they can accomplish a given task
- If it detects something is not working, it will be able to ask for help
- It should be able to assess the quality of its decisions
- If a cognitive robot is asked to perform a certain task, it should be able to say whether it can do it or not

Industrial Requirements for Cognitive Robots

7. Learning from Demonstration

- A cognitive robot should be able to learn new actions from demonstration by humans
- Instructions can be communicated through examples, including showing the robot the final results
  - The robot should be able to merge prior know-how and knowledge with learning by demonstration
  - Some of this prior knowledge should be extracted from codified organisational rules, regulations, and behavioral guidelines

Industrial Requirements for Cognitive Robots

7. Learning from Demonstration

- A cognitive robot should learn and adapt the parameters to achieve the task
- Teaching will exploit natural language, gaze and pointing gestures, and by showing the robot what to do and helping it when necessary
- When being taught, the robot should be anticipating what you are trying to teach it so that it predicts what you want it to do and then tries to do it effectively
- It should be possible to provide direct support for the robot, switching fluidly between full autonomy, partial autonomy, or manual control

Industrial Requirements for Cognitive Robots

8. Evaluating the Safety of Actions

- When it learns a new action, a cognitive robot should take steps to verify the safety of carrying out the action

Industrial Requirements for Cognitive Robots

9. Development and Self-Optimization

- A cognitive robots should develop and self-optimize
  - learning in an open-ended manner from its own actions and those of others (humans or other robots), continually improving its abilities
- It should be able to use what it has learned to determine possible ways to improve its performance, e.g. through internal simulation at times when the robot is not working on a given task

Industrial Requirements for Cognitive Robots

9. Development and Self-Optimization

- It should also be able to learn from its mistakes

Industrial Requirements for Cognitive Robots

10. Knowledge Transfer

- A cognitive robot should be able to transfer knowledge to other robots, even those having a different physical, kinematic, and dynamic configurations
- It should be able to operate seamlessly in an environment that is configured as an internet of things
    - The robot can be used, for example, as a way of collecting data from large experiments

Industrial Requirements for Cognitive Robots

11. Communicating Intentions and Collaborative Action

- A cognitive robot should be able to communicate their intentions to people around them
- Vice versa, it should be able to infer the intention of others, i.e. understanding what someone is doing and anticipating what they are about to do
- Ultimately, a cognitive robots should be able to collaborate with people on some joint task with a minimal amount of instruction

End of Lecture 1